

Water Segmentation using Multispectral and Optical Data (Second Week)

Objective:

Develop a robust solution for accurately segmenting water bodies using multispectral and optical data. This solution is vital for monitoring water resources, flood management, and environmental conservation, where precise segmentation can significantly impact decision-making

Tasks

- Research pretrained segmentation backbones suitable for multispectral data (e.g., ResNet, EfficientNet encoders).
- Adapt the first-layer input convolution to accept 12 channels instead of 3 (weight averaging or zero-init strategies).
- Use the segmentation-models-pytorch library to load a pretrained encoder + U-Net or DeepLabV3+ decoder.
- Fine-tune the pretrained model on the water segmentation dataset.
- Compare pretrained model performance against the scratch U-Net from Week 1 using the same metrics (IoU, F1, Precision, Recall).
- Document findings: which model performs better and why



Completion Requirements

- A pretrained segmentation model is successfully adapted to accept 12-channel input.
- The model is fine-tuned and achieves a measurably higher validation IoU than the Week 1 scratch model
- A comparison table of both models (IoU, F1, Precision, Recall) is included in the notebook/README.
- Code is clean, modular, and pushed to the GitHub repository

Resources

- <https://arxiv.org/abs/1606.00915>
- https://github.com/qubvel-org/segmentation_models.pytorch

